MASTER OF SCIENCE
IN ENGINEERING

# Week 4:
# Bayes and unbalanced classes

TSM_DeLearn

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# Preliminary discussion - assuming a classification system

Activity

- Why having unbalanced classes is bad?
- What can we do to fight this?

# Bayes and unbalanced classes

Bayes rule and examples

Unbalanced classes = prior probabilities

Implications for deep learning

Mister Bayes!

# Preamble - some examples

- Many classification problems are facing a problem of unbalanced classes:
  - Fraud prediction - Frauds are less frequent than genuine transactions
  - Natural disaster predictions - Disasters are much less frequent than normal events
  - Pathology detection - Few examples of healthy patient at training time while healthy patient are way more frequent at inference time

# Ideal classification : Bayesian approach

- **Bayes rule:** Elect as winning category the one having the largest *a posteriori* probability. Doing so we are guaranteed to maximise the accuracy.

likelihood
"probability of observing x given class j"

a priori probability
"probability of class j"

$$P(C_k|\mathbf{x}) = \frac{p(\mathbf{x}|C_k)P(C_k)}{p(\mathbf{x})}$$

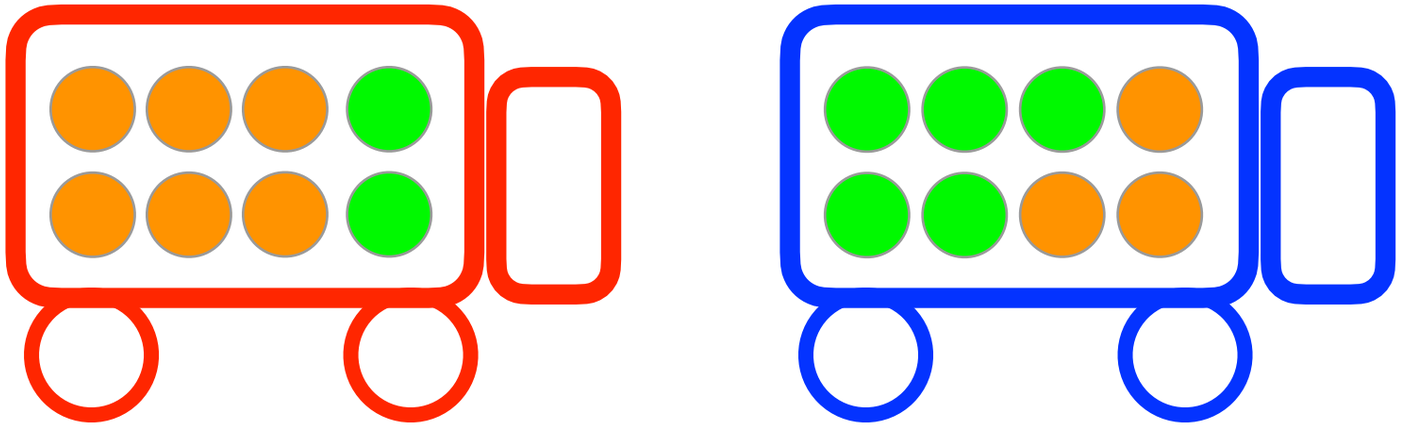
a posteriori probability
"probability of class j given observation x"

evidence = probability of x
"…unconditional to any class…"

- The *a priori* probability can be easily computed.
- The *likelihood* can be modelled with some *generative* statistical models such as histograms, Gaussian models (GM) or Gaussians mixture models (GMM).
- The *a posteriori* probability can also be modelled with some *discriminant* models

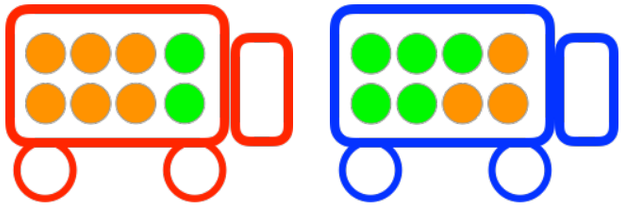
# Example 1 - discrete observations

- On a given road, there are red trucks and blue trucks. Both *class* of trucks are containing fruits.
- Let's *x* be the fruit type: orange / apple
- Let's the class be red trucks and blue trucks (trucks are containing fruits)



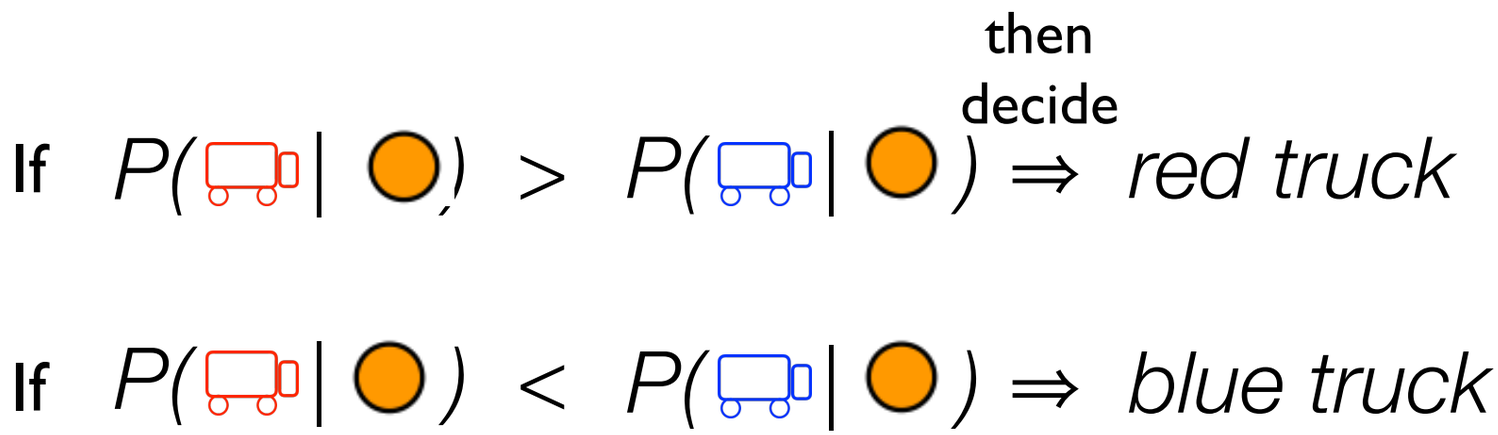
# Example 1 - discrete observations

- We observe *x* being an orange
- In which truck are we? *In which class are we?*
- Bayes'law tells us to look at the class having the largest a posteriori probability



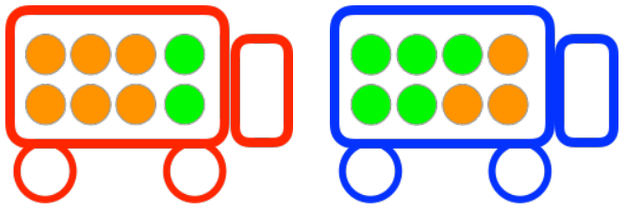
$$P(C_k|\mathbf{x})$$

a posteriori probability
"probability of class k given observation x"



# Example 1 - discrete observations

- We observe *x* being an orange
- In which truck are we?



*P*( | ) = 6/8 *P*( | ) = 3/8

likelihood
"probability of observing x given class j"

a priori probability
"probability of class j"

$$P(C_k|\mathbf{x}) = \frac{p(\mathbf{x}|C_k)P(C_k)}{p(\mathbf{x})}$$

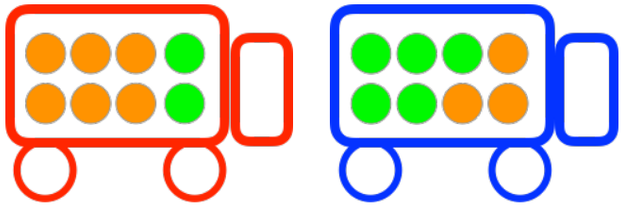
a posteriori probability
"probability of class j given observation x"

probability of x
"…unconditional to any class…"

The answer will depend to the apriori probability of having blue or red trucks, e.g. if we have many blue trucks, then this value will be high for the class "blue truck".

# Example 1 - discrete observations

- We observe *x* being an orange
- In which truck are we?
- Let's assume **equiprobable** trucks



$$P(C_{btruck}) = P(C_{rtruck}) = 1/2$$

$$P(C_k|\mathbf{x})$$

a posteriori probability
"probability of class j given observation x"

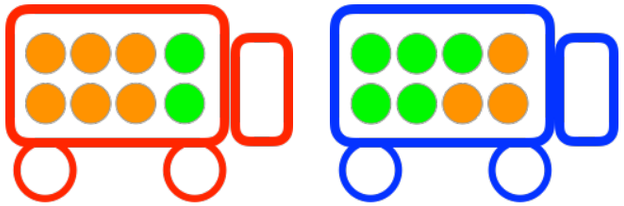
*P(* | *) = (6/8 .1/2) / p(x)*

*P(* | *) = (3/8 .1/2) / p(x)*

$\Rightarrow$ *red truck wins*

# Example 1 - discrete observations

- We observe *x* being an orange
- In which truck are we?
- Let's assume **blue trucks** are 80% of all trucks



$$P(C_{btruck}) = 0.8$$
$$P(C_{rtruck}) = 1 - 0.8 = 0.2$$

$$P(C_k|\mathbf{x})$$

a posteriori probability
"probability of class j given observation x"

*P(* *|* *) = (6/8 .0.2) / p(x)*

*P(* *|* *) = (3/8 .0.8) / p(x)*

$\Rightarrow$ *blue truck wins*

# Example 2 - women/men classifier with continuous observations

- Let's *x* be the length of hair

$C_1$ = Female
$N_1$ = 12 samples

$C_2$ = Male
$N_2$ = 20 samples

# Example 2 - women/men classifier with continuous observations

$P(C_1) = N_1/(N_1+N_2)$
$= 12/32 = 3/8$

$P(C_2) = N_2/(N_1+N_2)$
$= 20/32 = 5/8$

likelihood
"probability of observing x given class j"

a priori probability
"probability of class j"

$$P(C_k|\mathbf{x}) = \frac{p(\mathbf{x}|C_k)P(C_k)}{p(\mathbf{x})}$$

a posteriori probability
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probability of x
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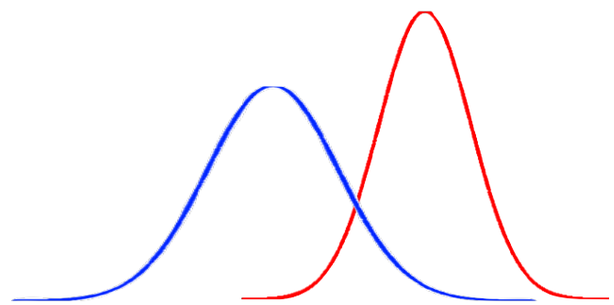
# Example 2 - women/men classifier with continuous observations



$P(x \mid C_1)$ $P(x \mid C_2)$

We may build "easily" estimators of likelihood, for example using histograms or make a hypothesis of a distribution and compute its parameters (for ex Gaussian distributions).

likelihood
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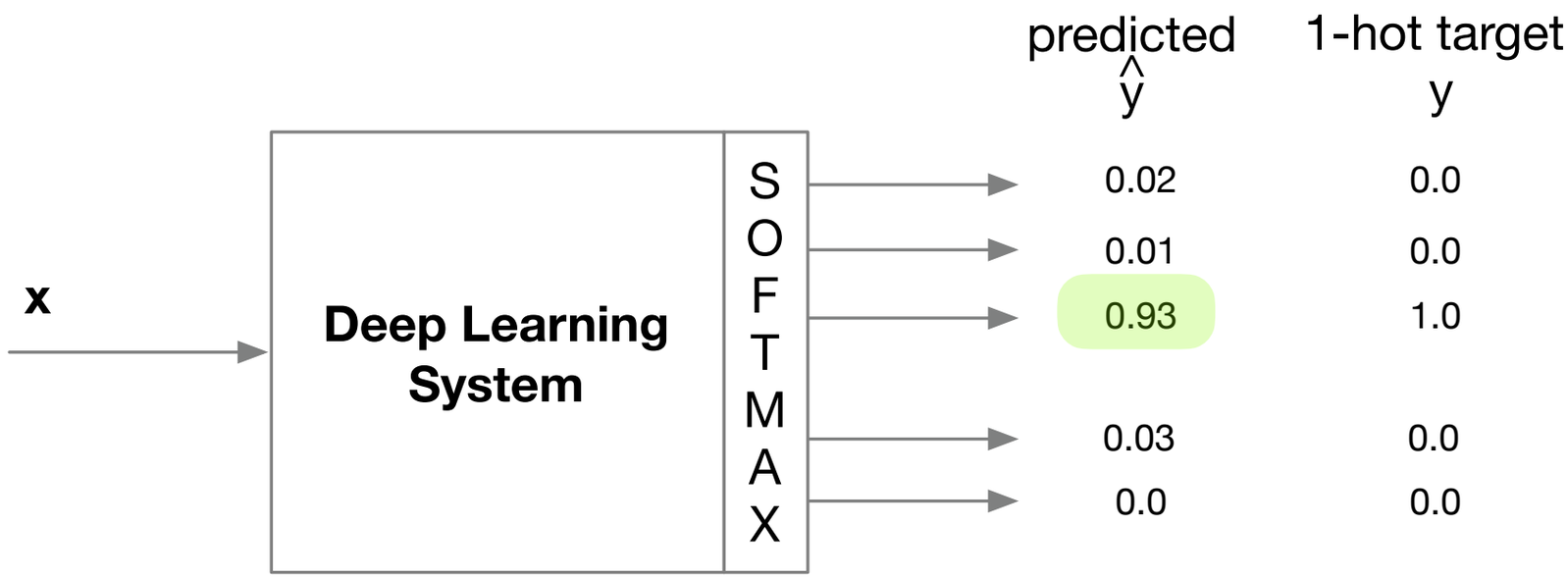
# Unbalanced classes have non equal prior probabilities

- The a priori probabilities $P(C_1)$, …, $P(C_k)$, …, $P(C_K)$ are actually the measures of the unbalances.
  - If they are all equal, it means we have a balanced training set
  - If they are unequal, the training set is unbalanced
- The a priori probabilities should be re-estimated whenever there is a change in the priors of the environment

A few samples of sharks

Many examples of tuna

# Link with Deep Learning Classification



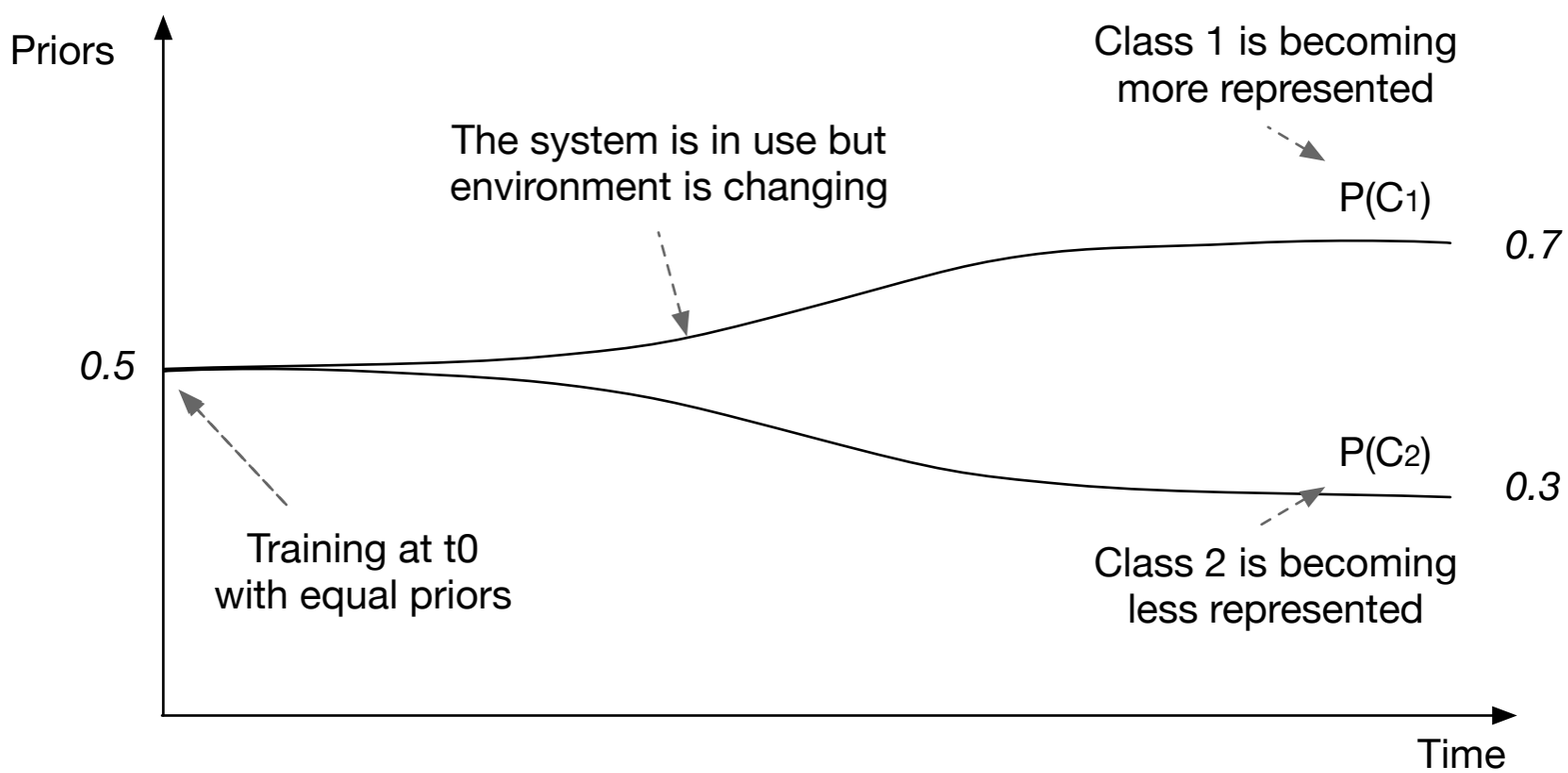

- The predicted $\hat{y}_k$ is actually an estimator of the posterior probability $P(C_k \,|\, \mathbf{x})$
- Bayes law tells us that this estimator includes the prior probability

$$P(C_k|\mathbf{x}) = \frac{p(\mathbf{x}|C_k)P(C_k)}{p(\mathbf{x})}$$

# Link with Deep Learning Classification

- Consequences of Bayes law for a trained system
  - The system will *favour* the most represented class, i.e. the one with the highest prior probability $P(C_k)$
  - If the priors are changing due to a change in the natural environment of the system:
    - We need to retrain with a training set reflecting the new priors
    - Or: we can use Bayes law to re-scale the posterior probability with the new priors

$$P(C_k|x) \leftarrow P(C_k|x)\frac{P(C_{k-New)}}{P(C_{k-Train})}$$



# How to handle an unbalanced training set?

A few samples of sharks

Many examples of tuna

- Strategy 1: do nothing, train as usual
  - Your classifier will include the priors, favouring *tuna* in this case (and you need to be aware of that!)
  - Re-scale system outputs to consider equal priors or other value of priors (see previous slide)
  - Warning: if too strong unbalances the classifier may just learn the priors
  - Note: un-balancing training data “on purpose” is actually a way to attack ML systems

# How to handle an unbalanced training set?

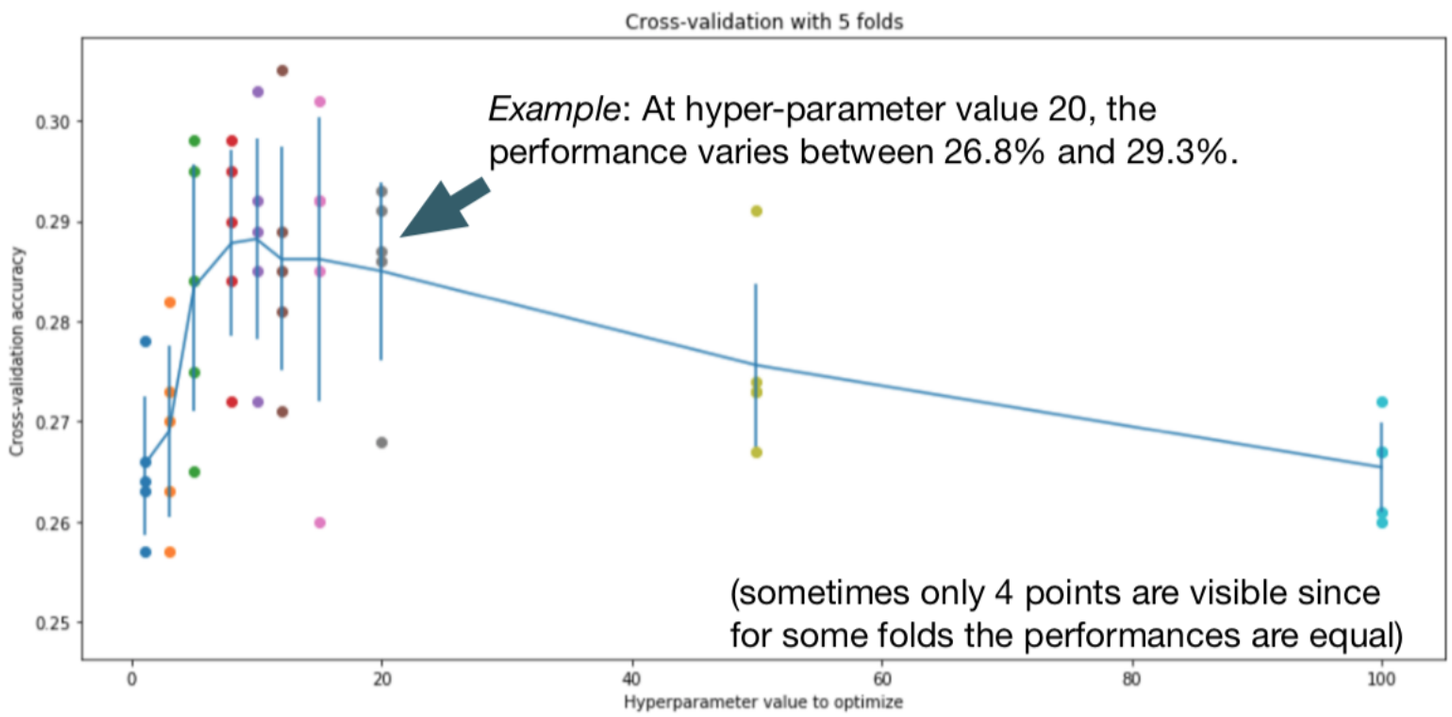
A few samples of sharks

Many examples of tuna

- Strategy 2: equilibrate your training set
  - Under-sample the most representative classes
    - Re-build a training set by taking *N* samples in each class, *N = min(N1, N2, …)*
    - Better: for each epoch, sample *N* samples in each class
  - Over-sample the less represented classes
    - Re-build a training set by repeating samples from the less represented classes
    - Better: use generative models or data augmentation

# A final note about test set size and unbalanced classes

- When unbalanced classes, it also means we have small numbers of test samples in the less represented classes.
  - It is really recommended to do **k-fold cross validation** (see Slides 28-29 of previous section)
  - we observe a variability of the predicted values when we do k-fold




**Example knn on cifar**

- "**The smaller the test size, the less confidence we should have in the claimed results**".
- There is a notion of interval of confidence behind any test!

# A final note about test set size and unbalanced classes

- The interval of confidence can actually be computed
  - E.g. see provided file on Moodle *confidence_intervals.xlsx* valid for two-classes problems
- Example: with a 300 test set size, an overall accuracy of 95% for a system should be reported as 95% ± 2%
  - i.e. we can't really say in this situation if a 95% system is really better than a 93% system

**CONDIFENCE INTERVAL OF AN ESTIMATED PROBABILITY**

| | |
|---|---|
| alpha | 0.05 |
| z | 1.96 |
| validity | 5 |

A **95% confidence interval** (alpha=0.05) is a range of values that you can be **95%** certain contains the true **mean** of the population.

| | Table 1 | | | | Table 2 (for comparison with Table 1) | | | |
|---|---|---|---|---|---|---|---|---|
| n | 1000 | | | | 300 | | | |
| 0.79 | 0.03 | 0.76 | 0.82 | 184.75 | 0.05 | 0.74 | 0.84 | 49.17 |
| 0.8 | 0.02 | 0.78 | 0.82 | 175.21 | 0.05 | 0.75 | 0.85 | 46.42 |
| 0.81 | 0.02 | 0.79 | 0.83 | 165.68 | 0.04 | 0.77 | 0.85 | 43.68 |
| 0.82 | 0.02 | 0.80 | 0.84 | 156.19 | 0.04 | 0.78 | 0.86 | 40.96 |
| 0.83 | 0.02 | 0.81 | 0.85 | 146.72 | 0.04 | 0.79 | 0.87 | 38.25 |
| 0.84 | 0.02 | 0.82 | 0.86 | 137.28 | 0.04 | 0.80 | 0.88 | 35.55 |
| 0.85 | 0.02 | 0.83 | 0.87 | 127.87 | 0.04 | 0.81 | 0.89 | 32.88 |
| 0.86 | 0.02 | 0.84 | 0.88 | 118.49 | 0.04 | 0.82 | 0.90 | 30.22 |
| 0.87 | 0.02 | 0.85 | 0.89 | 109.16 | 0.04 | 0.83 | 0.91 | 27.58 |
| 0.88 | 0.02 | 0.86 | 0.90 | 99.86 | 0.04 | 0.84 | 0.92 | 24.97 |
| 0.89 | 0.02 | 0.87 | 0.91 | 90.61 | 0.04 | 0.85 | 0.93 | 22.38 |
| 0.9 | 0.02 | 0.88 | 0.92 | 81.41 | 0.03 | 0.87 | 0.93 | 19.82 |
| 0.91 | 0.02 | 0.89 | 0.93 | 72.26 | 0.03 | 0.88 | 0.94 | 17.28 |
| 0.92 | 0.02 | 0.90 | 0.94 | 63.19 | 0.03 | 0.89 | 0.95 | 14.79 |
| 0.93 | 0.02 | 0.91 | 0.95 | 54.19 | 0.03 | 0.90 | 0.96 | 12.34 |
| 0.94 | 0.01 | 0.93 | 0.95 | 45.28 | 0.03 | 0.91 | 0.97 | 9.94 |
| 0.95 | 0.01 | 0.94 | 0.96 | 36.49 | 0.02 | 0.93 | 0.97 | 7.60 |
| 0.96 | 0.01 | 0.95 | 0.97 | 27.85 | 0.02 | 0.94 | 0.98 | 5.35 |
| 0.97 | 0.0106 | 0.9594 | 0.9806 | 19.43 | 0.02 | 0.95 | 0.99 | ***3.21*** |
| 0.98 | 0.0087 | 0.9713 | 0.9887 | 11.32 | 0.02 | 0.96 | 1.00 | ***1.25*** |
| 0.99 | 0.0062 | 0.9838 | 0.9962 | ***3.83*** | 0.0113 | 0.9787 | 1.0013 | ***-0.3778*** |
| 1.00 | 0.00 | 1.00 | 1.00 | ***0.00*** | 0.00 | 1.00 | 1.00 | ***0.00*** |

- Also, in case of unbalanced, reporting the global system accuracy is not recommended
  - Is a global accuracy of x% better than the highest prior value?
  - The confusion matrix and per-class metrics are recommended in such cases to make sure the system has learnt more than the priors.